Siddharth Sanjay

AI engineer

I build with Python, FastAPI, and React. At SkipCourse, I worked on RAG assessment workflows and asynchronous Google Cloud services.

GitHub ↗ LinkedIn ↗
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Education

Bachelor of Science in Computer Science, Arizona State University
May 2026

Experience

Software Engineering Intern

SkipCourse LLC
August 2025 – May 2026

- Refactored an AI assessment pipeline into a chunked RAG architecture using embeddings, vector similarity search, clustering, and LLM-based chunk evaluation.
- Integrated serverless microservices on Google Cloud with Cloud Functions Gen2, Firestore, Pub/Sub, and Vertex AI for asynchronous academic assessment and standards matching.

Experiential Learning Technology and Program Lead

University College, Arizona State University
January 2025 – May 2026

- Led creation, redesign, and maintenance of University College webpages serving 5,000+ users.
- Mentored interns in Drupal and web development, guiding webpage design, restructuring, and optimization.
- Coordinated outreach for Elevate and ELI through event planning, tabling, and stakeholder collaboration for 1000+ students.

Program Aide

University College, Arizona State University
January – December 2024

- Maintained University College websites in Drupal 8 and hosted information sessions promoting the Elevate platform.
- Collaborated with cross-functional teams on the ASU tutoring website's user experience and site architecture.

Featured projects

Harmonic Color Graph

A recommendation system that combines corpus-backed ranking, typed domain tools, and a grounded LLM assistant to explain chord choices.

Typed tools · Grounding · Evaluation · Python · FastAPI · Next.js · LangGraph · Claude · MCP · Recommendation ranking

ToneScope

A native audio workbench that connects signal measurements to an assistant workflow with revision-bound proposals, preview, explicit acceptance, and undo.

C++20 · Faust · React · JUCE · DSP · Native/UI integration · State validation · Numerical testing · Mutation testing

TruckerPath Dispatcher OS

A logistics prototype combining explainable load ranking with a Gemini PM Copilot that turns feedback into structured product hypotheses and experiment plans.

Next.js · Load scoring · PM Copilot · Tailwind CSS · Leaflet · Gemini API integration · API routes · Deterministic fallbacks · Weighted ranking

Skills in practice

- AI engineering: Python · LangGraph · Claude · Gemini · MCP · Typed tools · Grounded generation.
- Retrieval & data: RAG · Embeddings · Vector similarity search · Clustering · PostgreSQL · Redis.
- Cloud & pipelines: Google Cloud · Vertex AI · Cloud Functions Gen2 · Firestore · Pub/Sub · Asynchronous workflows.
- Evaluation & reliability: Schema validation · Intent evaluation · Latency measurement · Deterministic fallbacks · Rate limits · Cost controls · Audit records.
- Full-stack engineering: TypeScript · React · Next.js · FastAPI · API routes · Tailwind CSS · Native/UI integration.
- Recommendations & decisions: Feature-weighted ranking · Plausibility filtering · Diversity selection · Explainable scoring · Constraint handling.
- Audio systems: C++20 · JUCE · Faust · DSP · WAV rendering · Signal inspection · Aligned comparison.
- Verification & state control: Independent numerical evaluation · Adversarial testing · Mutation testing · Revision-bound proposals · Stale-state checks · Undo.
- Product: Workflow design · Scoring · Experiment planning · Feedback synthesis · Hypothesis development · PRD drafting.